Supplementary Information

Heliorhodopsin evolution is driven by photosensory promiscuity in monoderms

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We examined the distribution and co-occurrence of HeRs and type-1 rhodopsins (Supplementary Tables S4-S7 [https://doi.org/10.6084/m9.figshare.13286486]) in the GTDB database (release 89), as it has been previously suggested that these proteins tend to coexist within the same organisms[1]. From all 24,706 scanned genomes we identified and retrieved 1,455 *bona fide* type-1 rhodopsin-containing genomes from which only 69 genomes (4.74 %) proved to also harbour HeRs. Since only 15.33 % from all identified HeRs (n = 450) co-occur with type-1 rhodopsins we consider it as being more an exception than a norm and find no data to sustain a physiological dependency between these two rhodopsin families.

MORN-protein domain fusions

The bacteria encoding MORN-HeRs were previously predicted to be strict anaerobes[2]. Although mostly recovered from sediments, these MAGs also encode other proteins directly or indirectly associated with the presence of light (i.e. bacteriophytochrome COG4251, DNA repair photolyase COG1533, Deoxyribodipyrimidine photolyase COG0415). Therefore, the co-occurrence of strict anaerobiosis and light-dependent components indicates the top sediment layer as their likely habitat. The available 3D structure of MORN-repeats shows a single repeat to consist of short beta-pleated regions folded back upon themselves, creating a flat surface area that expands when the repeats are present in multiple tandem copies[3]. The presence of periplasmic proteins with MORN-Big_2 or MORN-Big2-PASTA domains may indicate the extracellular MORN repeats as adaptors between typical sensor domains (i.e. PASTA/Big_2) and the transducer protein kinases acting in the cytoplasm (Supplementary Figure S3). However, as the MORN-repeats that are fused to HeRs in these organisms are intracellular, they could only interact with MORN-repeats on the cytoplasmic side. In one such case, intracellular MORN-repeats were fused to ATPase component of an ABC-type efflux pump (likely involved in drug resistance or Cu2+/Na2+ ion efflux). The other candidate found was a PknB protein kinase-FHA-MORN repeat fusion that was predicted to be in an atypical membrane orientation. In such proteins, dimerization domains (e.g. PASTA) are extracellular (Supplementary Figure S3C) while cytoplasmic protein kinase domains

function as part of signal transduction pathways in a wide range of gram-positive bacteria[4]. The dimerization of PknB is essential for autophosphorylation and activation of the kinases through an allosteric mechanism[5]. The predicted reverse orientation (if correct) of this protein renders MORN-mediated interactions with HeR unlikely in these organisms. However, the presence of MORN-repeats in PknB type proteins for which dimerization is essential for function (most likely via MORN-repeats) strengthens the possibility that the MORN-repeats aid in MORN-HeR dimerization as well.

Other cysteine-rich motif-containing heliorhodopsins

Another extension found in four sequences (N-terminal variant 2 or ntv2 in Supplementary Table S8 [https://doi.org/10.6084/m9.figshare.13286486]) with many more cysteines (n = 10) that did not give any significant hits to known proteins was also identified. The presence of multiple cysteine residues and a conserved tryptophan residue is reminiscent of RINGv finger domains that coordinate two metals. However, RING finger domains also have a highly conserved histidine residue that was not detected. Indeed, we also found at least two instances in freshwater Actinobacteria (*Ca. Planktophila*) where a RINGv domain-containing protein is located upstream of the heliorhodopsin gene and in the same orientation (see Figure 3).

A third variant (ntv3) with seven cysteines was found (in Thermoplasmatales and in Euryarchaeota) but without conserved histidines or tryptophan. However, it shows broad similarity with ntv3 in presence of conserved prolines and arginines before the cysteine motifs. Apart from the N-terminal extensions motifs, at least three sequences of the intracellular loop (ICL3) were also found to be rich in cysteines (n = 8) which also presented a conserved tryptophan similar to ntv2 described above (intracellular loop 3 variant 1, icl3v1).

Thus, at least 17 sequences presented cysteine-rich motifs either at the N-terminus (ntv1, ntv2 and ntv3) or in the intracellular loop (icl3v1) at the cytoplasmic side of heliorhodopsins suggesting the possibility that these might be transducers of the conformational change in heliorhodopsins upon light excitation. The conserved cysteines in these proteins could bind

either iron or zinc and are likely redox-active. While we did find many other types of N-terminal extensions, we were unable to find any significant hits to these even by sensitive sequence searches (Supplementary Table S8 [https://doi.org/10.6084/m9.figshare.13286486]).

Sources of retinal for HeR function

We also found several NAD-dependent short-chain dehydrogenase genes that might encode for retinol dehydrogenases in the vicinity of HeRs (adh_short in Figure 3, Supplementary Table S9 [https://doi.org/10.6084/m9.figshare.13286486]). It has been mentioned before that as HeRs are able to efficiently capture retinal from exogenous sources and that HeR-encoding microbes do not have a retinal biosynthesis pathway[6].

De novo retinal biosynthesis requires five genes that if supplied in-trans to a non-retinal producing microbe may result in functional rhodopsins[7]. The final step of the *de novo* pathway uses a beta-carotene monooxygenase that converts beta-carotene to retinal. However, retinal may also be converted from retinol by the action of retinol dehydrogenases. We used the curated GTDB database to further probe the co-occurrence of HeR and type-1 rhodopsins along with genes for retinal biosynthesis. Of the total of 381 genomes that encoded only HeR, we find that only a single genome encoded all genes necessary for the *de novo* production of retinal, but 213 (55%) also encoded the beta-carotene monooxygenase and 241 genomes (63%) encoded at least one retinol dehydrogenase (Supplementary Table S10 [https://doi.org/10.6084/m9.figshare.13286486]). Considering genomes that encoded only type-1 rhodopsins (n = 1,386), 596 (43%) encoded the complete pathway for retinal biosynthesis and additionally 995 (71%) also encoded at least one retinol dehydrogenase. It appears that microbes encoding only HeR mostly lack the complete pathway for *de novo* retinal biosynthesis and that apart from exogenous capture of retinal, conversion from beta-carotene (via beta-carotene monooxygenase) or from retinol (via retinol dehydrogenases) may be at work.

HeR genomic context

We performed gene context analysis of HeRs by combining the maximum-likelihood phylogenetic tree generated for representative HeR sequences (n = 872) with HeR gene neighbourhood information (Supplementary Figure S4; iTOL: https://itol.embl.de/tree/14723125092414101632755822). The resulting tree places most HeR sequences (n = 835) within 19 conspicuous phylogenetic clusters which we further denominate as C1-C19 (see Supplementary Figure S15). Among them, Actinobacteriota-encoded HeRs are by far the most numerous (n = 533) accounting for eight well-defined clusters (i.e. C1-7 and C11) and a small sub-cluster within Patescibacteria (CPR)-dominated C17. Regarding Actinobacteriota, C1 and C3 are represented by order Nanopelagicales, C2 includes chiefly members of Microtrichales (class Acidimicrobiia), C4 comprises Actinomycetales HeRs and C5 includes classes Coriobacteriia and Thermoleophilia. Notably, C5 brings together HeRs recovered mainly from lesser studied sediment habitats including nine Chloroflexota (class Dehalococcoidia) sequences. Predominantly marine C6 includes Acidimicrobiia HeRs from order Microtrichales and other poorly classified representatives from within this class while C7 has both marine and freshwater Microtrichales together with Propionibacteriales genus *Nocardioides*.

Cluster C11 stands out in this analysis due to the high level of evolutionary conservation of both HeRs and their neighbouring genes, bringing together exclusively members of marine Actinobacteria from order *Ca.* Actinomarinales (TMED189). This genomic region has been discussed before [1] and the proximity of HeRs to *nuo* genes was interpreted as a clue towards involvement of HeRs in the electron-transfer chain. Importantly, in C11 we notice that synteny is only conserved among genes sharing the same orientation as HeR while gene “gains” and/or “losses” occur only in the opposite orientation. Despite very high phylogenetic relatedness within C11 and therefore the unsurprisingly similar gene context amongst its members, the differences between (+) and (-) strand feature conservation (relative to HeR orientation) indicate a potentially relevant transcriptional unit comprised of

genes: afuA - iron(III) transport system substrate-binding protein (K02012), *afuB* - iron(III) transport system permease protein (K02011), *afuC* - iron(III) transport system ATP-binding protein (K02010), *HeR* – Heliorhodopsin (PF18761), an 11-subunit respiratory complex I operon (*nuoA, B, C, D, H, I, J, K, L, M, N*), *NDUFAF7* - NADH dehydrogenase [ubiquinone] 1 alpha subcomplex assembly factor 7 (K18164/PF02636), *htpX* - heat shock protein (K03799), *pspE* - phage shock protein E (K03972) containing a *Rhodanese* (PF00581) domain and *adenosine_kinase* (cd01168). In summary, HeRs from C11 are always preceded by genes encoding a complete ABC-type ferric iron uptake system and followed by an operon encoding an 11-subunit, “ancestral”-type[8] respiratory complex I and by accessory components required for correct assembly and function of this complex (*NDUFAF7, htpX, pspE*) [9–12]. The last conserved gene encodes a pfkB family adenosine kinase (cd01168), a key purine salvage enzyme that phosphorylates adenosine to generate adenosine monophosphate (AMP) [13].

The presence of HeRs within the same transcriptional unit as above-mentioned energy metabolism components could indicate them as modulators or even light-induced sensory “switches” of such processes, a mechanism perhaps similar to cryptochrome-driven metabolic synchronization with substrate availability described in other Actinobacteria[14]. Notably, beside the 11-subunit complex I, which lacks the NADH dehydrogenase module (subunits nuoE, nuoF, nuoG) [8], *Ca. Actinomarina* (TMED189) genomes also encode the full-sized 14-subunit variant of respiratory complex I in close proximity to the first (for example in GCA-902516125.1). Curiously, the association of HeRs with complex I genes is reminiscent of that between the transmembrane, sensory, EAL-domain containing protein seen in *Bacillus cereus* located upstream of a similar 11-subunit complex I operon[8].
The last cluster featuring a significant number of Actinobacteria HeRs (n = 12) is C17. In this CPR-dominated cluster, Actinobacteria HeRs form a well-defined group sharing a common ancestor with a small CPR sub-cluster. While gene context appears conserved within these Actinobacteria, this does not apply to the putative “sister” CPR sub-cluster.

Clusters C8-C10 share a common ancestor with C11 and include HeRs encoded largely in strict or facultative anaerobic prokaryotes recovered from sediments (including activated sludge). Notably, the phylogenetic tree (iTOL link above) shows two Asgardarchaeota (class Heimdallarchaeia) HeRs branching with very high support (SH-test/UFBoot = 97.2/98) as a sister clade to all C8-C10 members, after the split with the common ancestor shared with C11. Despite the high support for the Asgardarchaeota HeR split, defining a credible cluster will require including additional sequences once more genomes become available. The basal C10 cluster comprises Archaea-derived HeRs from phyla Crenarchaeota (class Bathyarchaeia) and Euryarchaeota (class Methanobacteria). Clusters C8 and mainly C9 include the MORN-HeR encoding Firmicutes as well as a few Chloroflexota (class Anaerolineae) HeRs.

C12-C16 form a separate super-cluster showing moderate-to-low support for internal branching patterns and include mostly, although not exclusively, anaerobic Archaea (C12 and C14, with the notable exception of aerobic Halobacterota within C12) and anaerobic Chloroflexota (classes Anaerolineae - C13, C15 and Dehalococcoidia – C16). Notably, C16 includes one Thermoplasmatota HeR (encoded in contig SRR5506739-C331) with a zinc-finger extension (Znf-HeR) at the N-terminus. C18 is the most basal cluster with confidently assigned taxonomy. It includes exclusively aerobic Chloroflexota members of the Ellin6529 lineage.

Although no consensus taxonomy could be determined for members of C19 – the first branching group after the split with proteorhodopsins, the abundance of “eukaryotic signature proteins” (ESP) (e.g. Arf, Roc, Rab, etc.) points towards either a eukaryotic origin or unidentified, ESP-rich archaea[15,16].

An extended phylogenetic tree of HeRs, including additional dereplicated sequences identified and retrieved from UniProtKB and GTDB, is available in FigShare (see Supplementary Methods - Phylogenetic tree of HeRs).

Supplementary Methods

Re-assembling of HeR-encoding Spirochaeta. The unexpected detection of HeR in a previously published *Spirochaeta* (diderm organism) genome prompted further investigation. The original Illumina short-read dataset SRX2623364 was downloaded from NCBI SRA (Sequence Read Archive) and preprocessed by using a combination of tools provided by the BBMap project (https://sourceforge.net/projects/bbmap/). This involved removing poor-quality reads with bbduk.sh (qtrim = rl, trimq = 18), identifying phiX and p-Fosil2 control reads (k = 21) and removing Illumina sequencing adapters (k = 21). Further, *de novo* assembly of preprocessed paired-end reads was done by Megahit v1.2.9 [17] with k-mer list: 29, 39, 49, 59, 69, 79, 89, 99, 109, 119, 127, and with default parameters. A total of 886 contigs with an average length of 4.98 kbp were produced. Protein-coding genes were predicted by Prodigal[18] and taxonomically classified by scanning with MMSeqs2 against the GTDB database. A *Spirochaeta* contig (length = 304,032 bp) was identified and scanned for the presence of HeR against the PFAM database. Both taxonomy (*Spirochaeta*) and HeR presence were consistent with previously published results.

Phylogenetic tree of HeRs. An extensive collection of predicted HeR amino acid sequences (n = 4,108) was generated from: 1) all HeR sequences available in UniProtKB, 2) HeR identified within dereplicated contigs (using cd-hit-est -c 0.95 -aS 0.95) assembled from publicly available metagenomes and metatranscriptomes, 3) HeR identified within dereplicated high-quality MAGs included in the GTDB database and 4) HeR assembled from the strand-specific metatranscriptomic dataset generated in this study. This collection was simplified by keeping only representative sequences (n = 1,669) chosen following clustering with MMSeqs2[19] at 90% sequence identity and 90% coverage (mode: easy-cluster; -c 0.90; --min-seq-id 0.90). Representative HeR sequences were aligned together with 30 selected proteorhodopsins serving as outgroup by using PASTA[20] (resulting alignment with 1,699 sequences, 2,486 columns, 2,264 distinct patterns, 1,091 parsimony-informative sites, 698 singleton sites, 697 constant sites). A Maximum Likelihood (ML) phylogenetic tree was

constructed with IQ-TREE2 [21] (1,000 iterations for ultrafast bootstrapping[22] and SH testing, respectively; best model chosen by ModelFinder[23]: LG+G4; additional parameters recommended for short sequence alignments -nstop 500 -pers 0.2). The generated tree was annotated to include labels containing: HeR-encoding contig name, habitat of origin and consensus GTDB taxonomic classification (if available). Data including alignment and the annotated phylogenetic tree are deposited in FigShare (https://doi.org/10.6084/m9.figshare.13286486).

Phylogenetic tree of HeRs with gene context. A simplified depiction of HeR genomic context across representative taxonomic groups harbouring such genes was constructed by merging HeR phylogenetic information with available HeR gene neighbourhood data (Supplementary Figure S4). For this purpose, we established a collection of representative, dereplicated HeR-encoding contigs (n = 872) of at least 5 kb and with clear consensus taxonomy from two sources: 1) HeR-encoding contigs assembled from publicly available metagenomes and metatranscriptomes (n = 3,145) and 2) HeR contigs assembled from the strand-specific metatranscriptomic dataset generated in this study (n = 6). Dereplication of contigs was previously achieved using cd-hit-est[24] with identity cutoffs of 95% and coverage of 95% (-c 0.95 -aS 0.95). **Phylogenetic tree building:** Curated HeR sequences recovered from selected contigs were aligned together with 30 proteorhodopsins serving as outgroup by using PASTA[20] (resulting alignment with 902 sequences with 957 columns, 895 distinct patterns, 550 parsimony-informative sites, 198 singleton sites, 209 constant sites). A Maximum Likelihood (ML) phylogenetic tree was constructed with IQ-TREE2[21] (1,000 iterations for ultrafast bootstrapping[22] and SH testing, respectively; best model chosen by ModelFinder[23]: LG+I+G4; additional parameters recommended for short sequence alignments -nstop 500 -pers 0.2).

Reconstruction of HeR gene neighborhoods: Coding sequences were predicted *de novo* using Prodigal[18] in metagenomic (-p meta) mode. Protein domains were annotated by scanning predicted coding sequences against the PFAM (Protein Families) database using

the publicly available perl script pfam_scan.pl. Predicted protein sequences from all contigs were clustered together using the MMSeqs2 easy-cluster workflow with 50% identity and 80% coverage cutoffs (--min-seq-id 0.5 -c 0.8 -e 1e-3 --cluster-reassign) and a minimum of 2 sequences per cluster. Clusters were sorted according to their size (i.e. number of sequences) and colour codes were assigned to the top largest 63. All clusters containing HeRs were assigned matching colours. HeR gene neighbourhood data was combined with the reconstructed HeR phylogenetic tree in iTOL[25] (https://itol.embl.de/tree/14723125092414101632755822). To facilitate visual interpretation, a number of adjustments and rules were applied: 1) all contigs were oriented according to the sense (+) of encoded HeRs, 2) contigs were centered on HeR genes with a maximum of 10 neighbouring genes depicted up- and downstream, 3) information regarding gene lengths was not included, all of them being shown as equally sized rectangles, 4) homologous genes (i.e. members of the same MMSeqs2 defined cluster) share matching colours within each phylogenetically defined cluster, 5) all HeRs are coloured the same across all phylogenetic clusters, 6) grey rectangles indicate genes with few homologues and/or singletons, 7) taxonomy, habitat information and phylogenetic clusters are colour coded on independent strips.

MORN-HeR phylogenomic tree. A maximum-likelihood (ML) phylogenomic tree was constructed for Firmicutes MAGs encoding MORN-HeR protein domain fusions along with other representatives of this phylum that presumably lack such genes. The established collection of (n = 68) MAGs and reference genomes (Supplementary Table S15 [https://doi.org/10.6084/m9.figshare.13286486]) was scanned by hmmsearch against a previously published list of (n = 120) conserved protein marker HMMs[26]. Four divergent markers (TIGR00442, TIGR00539, TIGR00643, TIGR00717) were identified by scanning with CD-search (e-value < 1e-2) and removed. Curated amino acid sequences for the selected 116 phylogenetic markers were aligned with PRANK[27] and resulting alignments were trimmed by BMGE (parameters: -t AA -g 0.5 -b 3 -m BLOSUM30) [28]. Individually

trimmed alignments were concatenated using the script catfasta2phyml.pl (available from https://github.com/nylander/catfasta2phyml) resulting in a block of 68 sequences with 40,714 columns, 36,088 distinct patterns, 28,978 parsimony informative sites, 3,366 singleton sites and 8,370 constant sites. The concatenated alignment was used with IQ-TREE (v.1.6.12) to construct the ML phylogenomic tree (parameters: 1,000 iterations of ultrafast bootstrapping[22] and SH testing[21], respectively; best model (LG+F+R5) chosen by ModelFinder[23].

Multiple sequence alignment (MSA) of MORN-HeR proteins. Predicted amino acid sequences containing full-length MORN-MORN-MORN-HeR protein domain fusions (n = 36) were retrieved from Firmicutes MAGs (n = 35) previously used to construct the phylogenomic tree presented in Supplementary Figure S1. All sequences were aligned together using the PSI-Coffee alignment method[29] provided by the T-Coffee online server (http://tcoffee.crg.cat) with default parameters. The resulting MSA is presented with annotations in Supplementary Figure S2 while the original alignment file generated by PSI-Coffee is available in FigShare (https://doi.org/10.6084/m9.figshare.13286486).

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